

Exercise 2 - Results

Question 1. Answer:

- (a) 00100011
- (b) 11101000
- (c) Require 9 bits at least
- (d) 10000010

Question 2. Answer:

- (a) 13
- (b) -103
- (c) -1
- (d) 99

Question 3. Answer:

- (a) 11101
- (b) 101.111
- (c) 011111
- (d) 111111

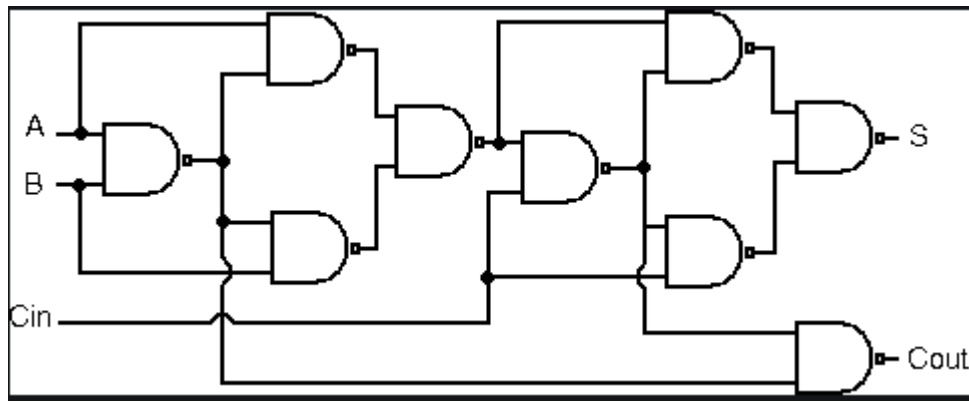
Question 4. Answer:

- (a) 58_{BCD}
- (b) 60_{BCD}
- (c) 1110_{BCD}
- (d) 1583_{BCD}

Question 5. Answer: 0xF022DBB3

Question 6. Answer: Overflow = $\overline{x \oplus y} \ x \oplus z$

Question 7. Answer:



Question 8. Answer:

CLA:

- The carry bits are generated for all the stages of the addition at the same time as soon as the input signal is provided.
- Propagation delay is lower.
- More resource usage for implementation.
- Recommended for large number of computations.

RCA:

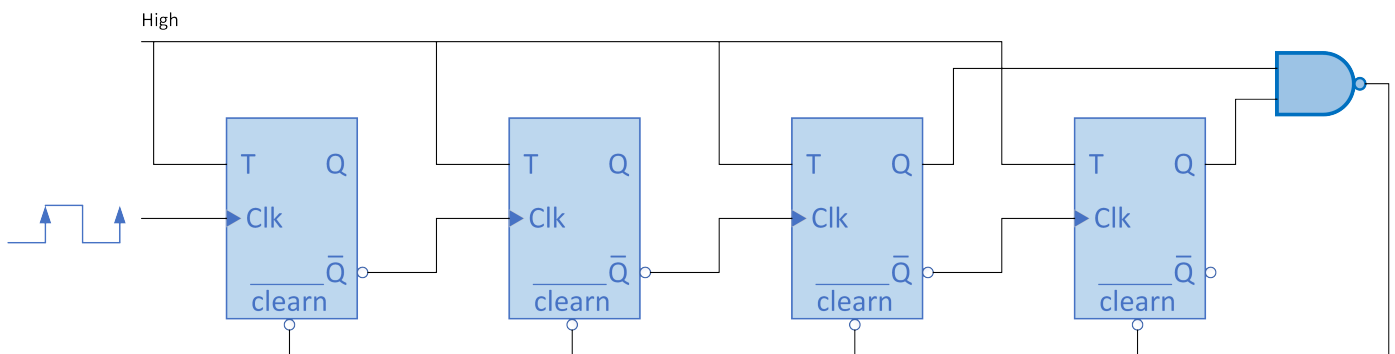
- The carry bit ripple through all the stages of the adder.
- Propagation delay is higher.
- Less resource usage for implementation.
- Recommended for infrequent computation tasks.

(a) 2048

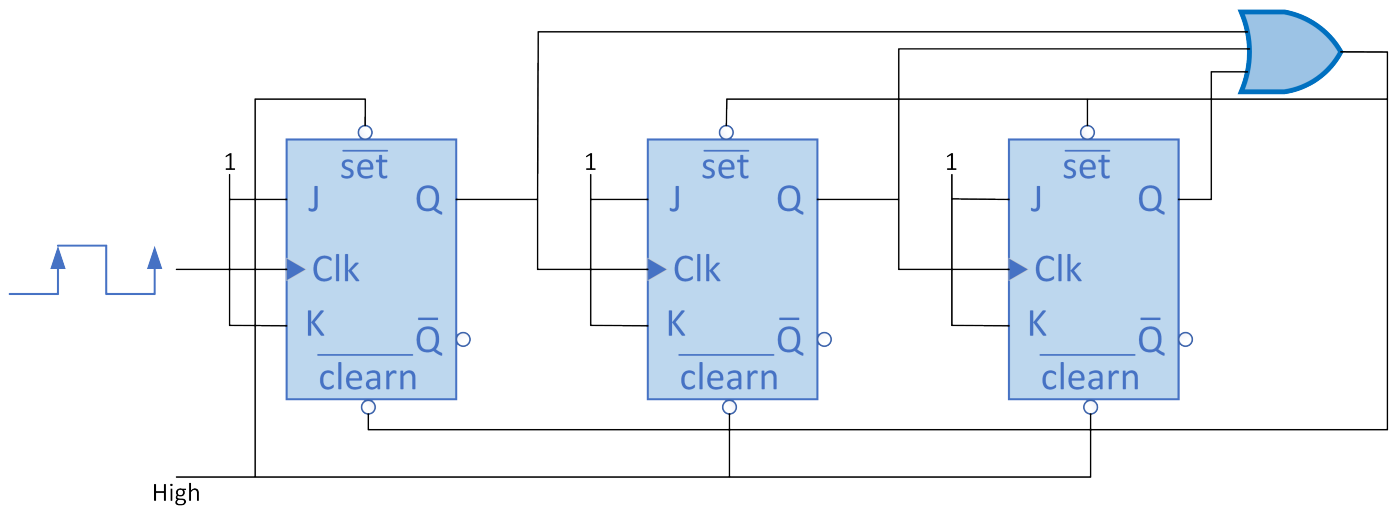
(b) $0 \rightarrow 2047$

Question 9. Answer:

Question 10. Answer:



Question 11. Answer:



Question 12. Answer:

- (a) 12
- (b) Q_C
- (c) $1/2$ (A), $1/2$ (B), $2/3$ (C), $2/3$ (D)

Question 13. Answer: Assume A is MSB, D is LSB

$$\begin{aligned} J_A &= BCD \\ K_A &= D \\ J_B &= CD \\ K_B &= CD \\ J_C &= A'D \\ K_C &= D \\ J_D &= 1 \\ K_D &= 1 \end{aligned}$$

Question 14. Answer: Assume A is MSB, C is LSB

$$\begin{aligned} D_A &= BC' + AB + A'B'C \\ D_B &= A'B' + B'C' \\ D_C &= A' + B + C' \end{aligned}$$

Question 15. Answer:

$$\begin{aligned} F_A &= a'b'A \\ F_B &= a'bB \\ F_C &= ab'C \\ F_D &= abD \\ F &= F_A + F_B + F_C + F_D \end{aligned}$$

Question 16. Answer:

$$\begin{aligned} A &= a'b'F \\ B &= a'bF \\ C &= ab'F \\ D &= abF \end{aligned}$$

Question 17. Answer:

For D

$$I_0 = Y$$

$$I_1 = Y'$$

$$\text{Select} = X$$

For B

$$I_0 = Y$$

$$I_1 = 0$$

$$\text{Select} = X$$

Question 18. Answer:

$$Y(D, C, B, A) = \sum m(3, 11, 12, 14)$$

Question 19. Answer: 10

Question 20. Answer:

Input : 7

Output : 128

Question 21. Answer:

Assume

Input : $D_7(\text{MSB}) \rightarrow D_0$

Output : $Y_2(\text{MSB}) \rightarrow Y_0$

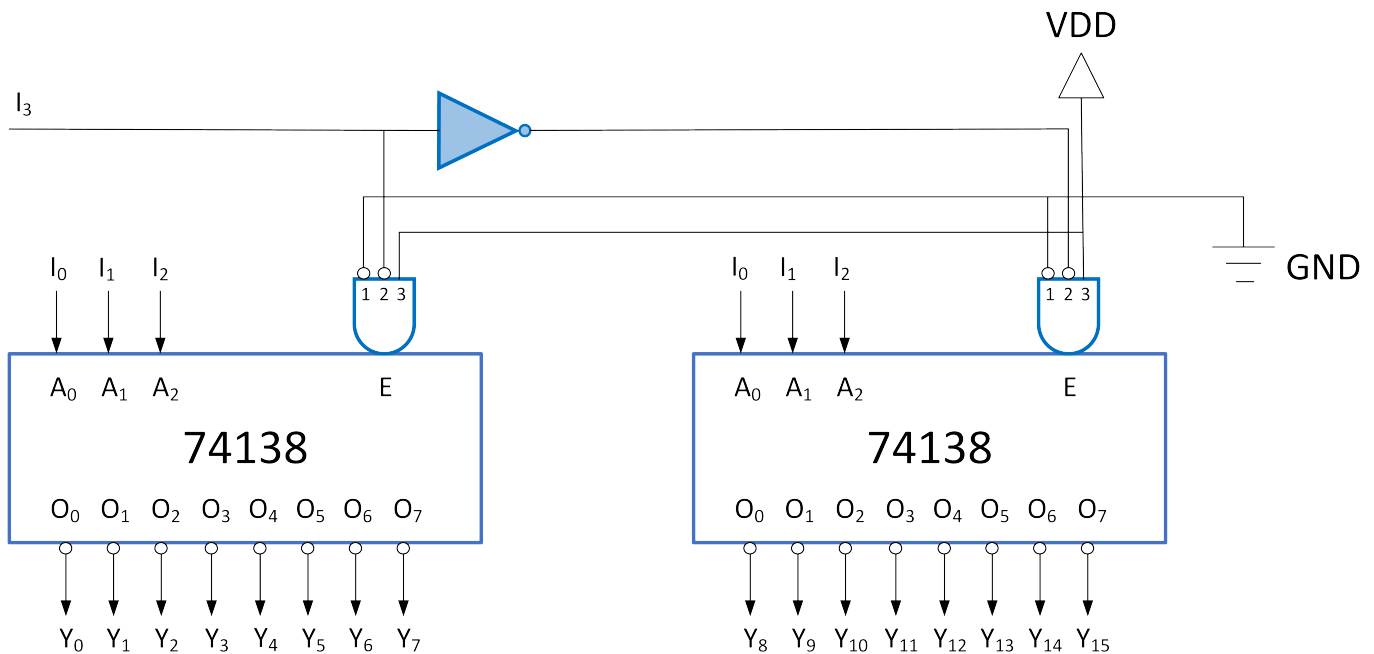
Then

$$Y_2(\text{MSB}) = D_4 + D_5 + D_6 + D_7$$

$$Y_1(\text{MSB}) = D_2 + D_3 + D_6 + D_7$$

$$Y_0(\text{MSB}) = D_1 + D_3 + D_5 + D_7$$

Question 22. Answer:



Question 23. Answer:

Question 24. Answer:

